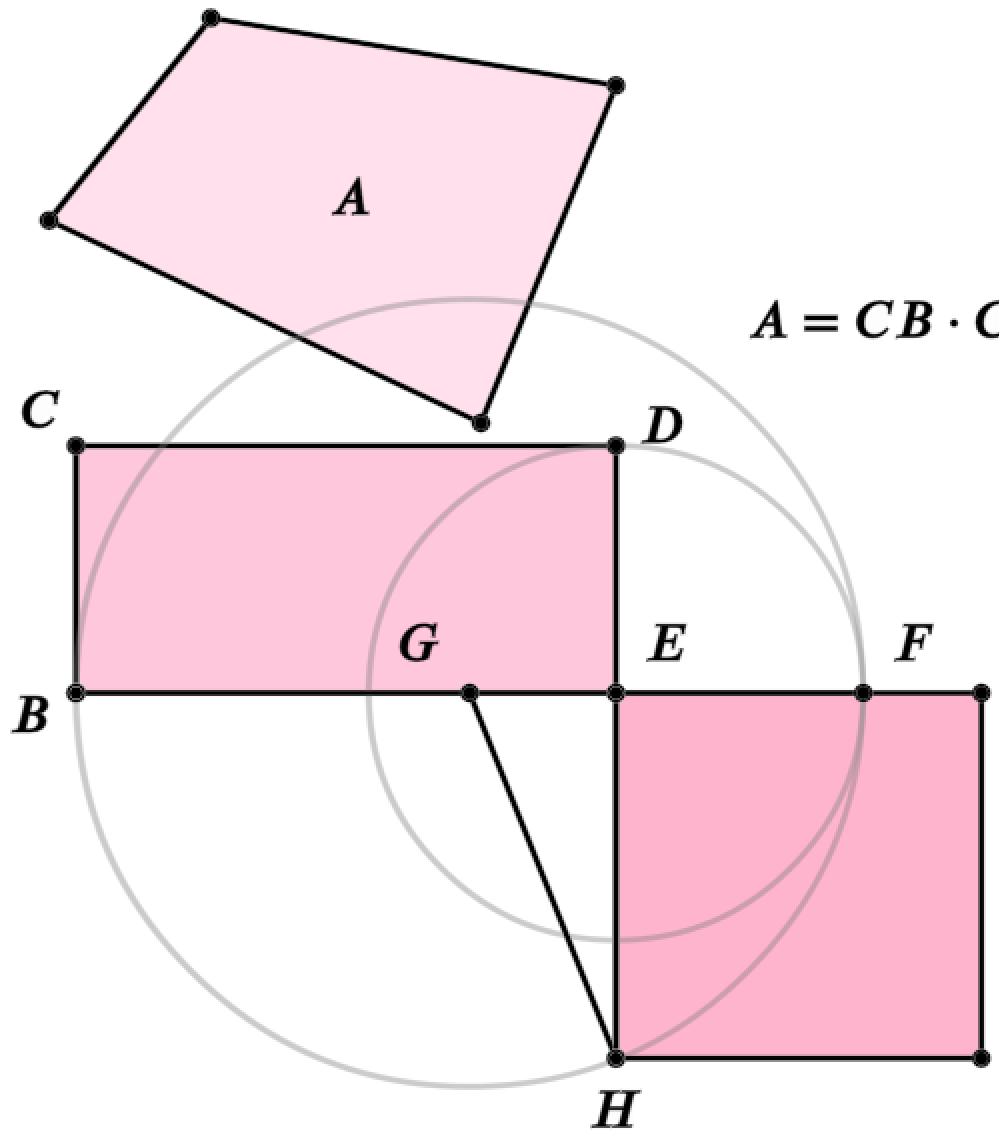
Euclid's Elements

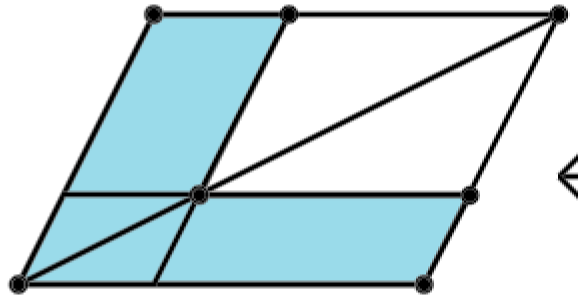
Book II



$$A = CB \cdot CD = (EH)^2$$

It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)



Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

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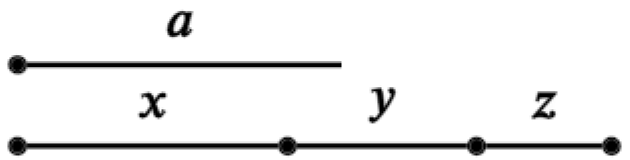
1. 

$a(x+y+z) = ax + ay + az$

2. 

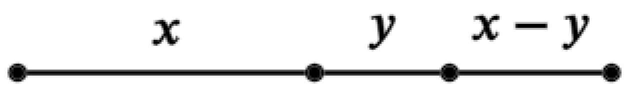
$(x+y)^2 = x(x+y) + y(x+y)$

3. 

$y(x+y) = yx + y^2$

4. 

$(x+y)^2 = x^2 + y^2 + 2xy$

5. 

$(x+y)(x-y) + y^2 = x^2$

6. 

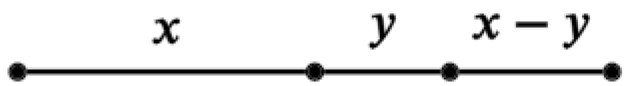
$y(2x+y) + x^2 = (x+y)^2$

7. 

$(x+y)^2 + y^2 = x^2 + 2y(x+y)$

8. 

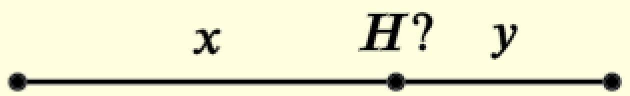
$(x+2y)^2 = 4y(x+y) + x^2$

9. 

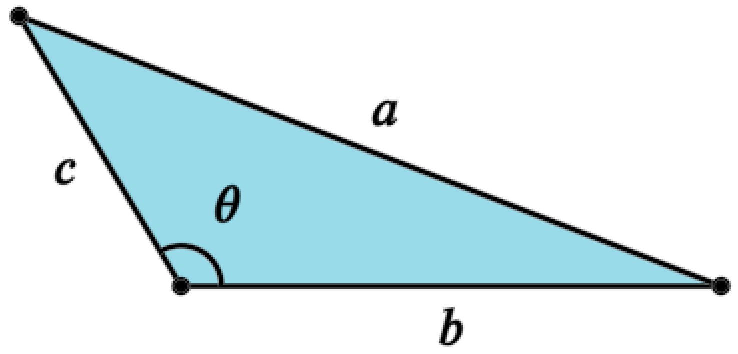
$(x+y)^2 + (x-y)^2 = 2(x^2+y^2)$

10. 

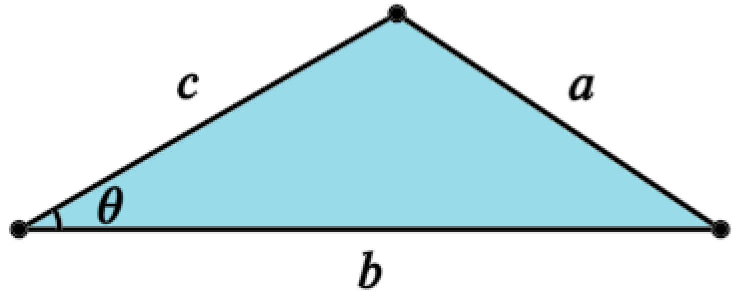
$(2x+y)^2 + y^2 = 2(x^2 + (x+y)^2)$

11. 

Find H such that: $y(x+y) = x^2$

12. 

Cosine Law, $a^2 = b^2 + c^2 - 2bc\cos\theta$

13. 

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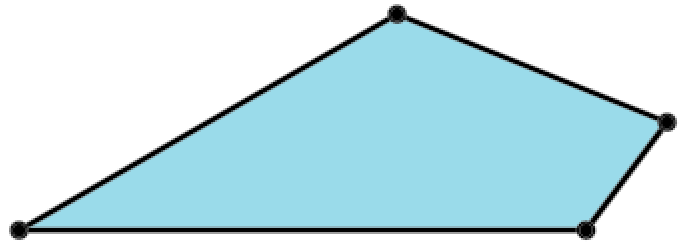
14. 

Construct a square equal to the polygon

Proposition 11 of Book 2

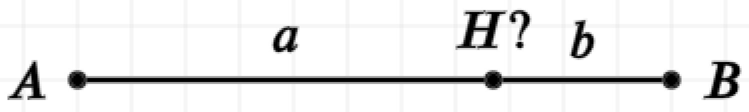
To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

$$AB \cdot BH = AH^2$$

$$(a + b)\, b = a^2$$



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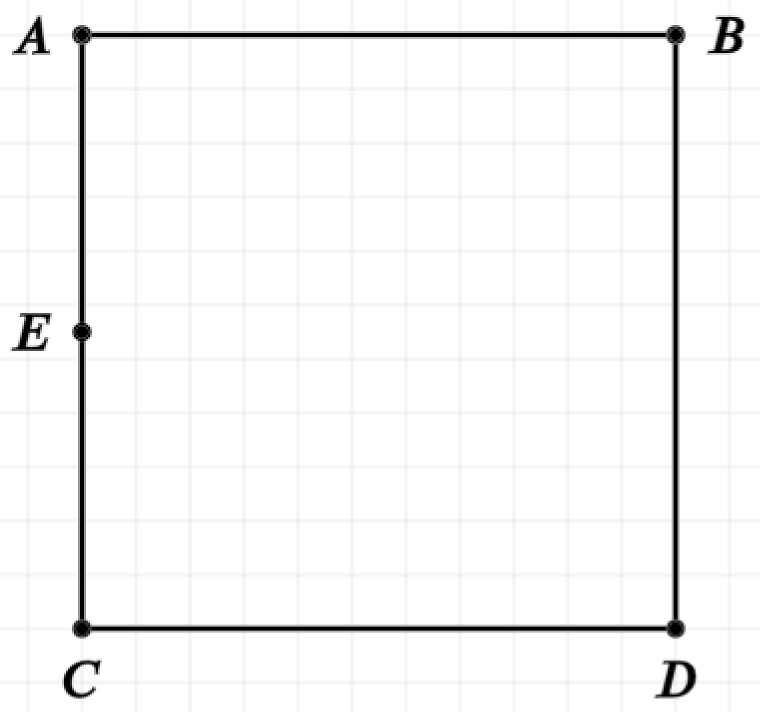

$AE = EC$

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

Construction

Draw a square ABCD on AB (I.46) and bisect AC (I.10) at point E

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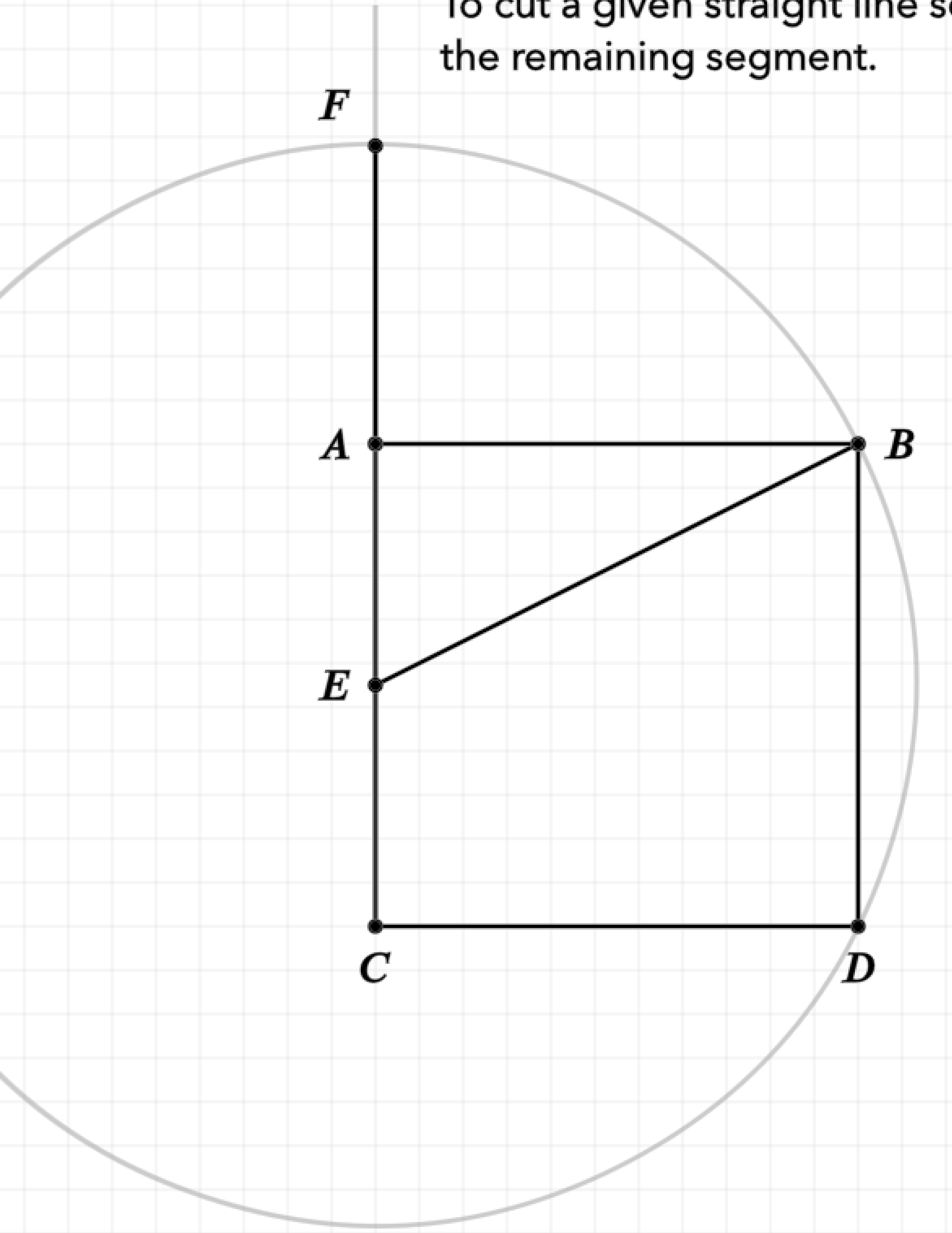


$AE = EC$

$EF = EB$

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

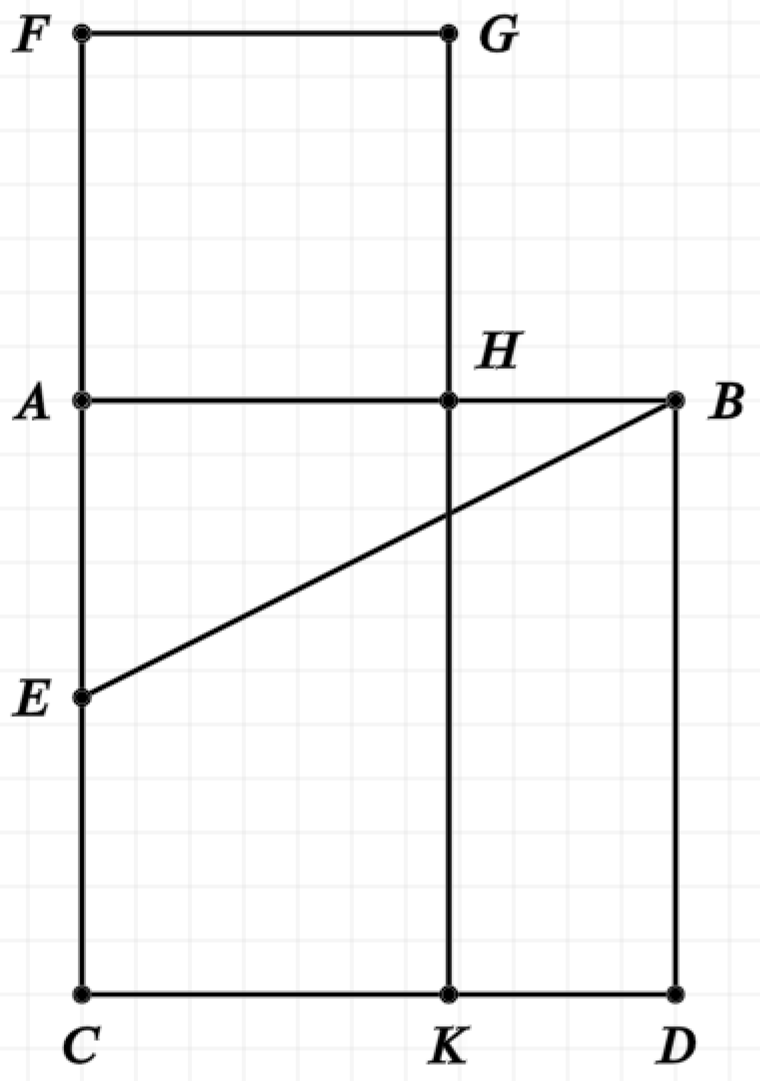
Construction

Draw a square ABCD on AB (I.46) and bisect AC (I.10) at point E

Let EB be joined, and extend CA to F such that EF equals AB

Proposition 11 of Book 2

To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.



$AE = EC$
$EF = EB$
$FA = AH$

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

Construction

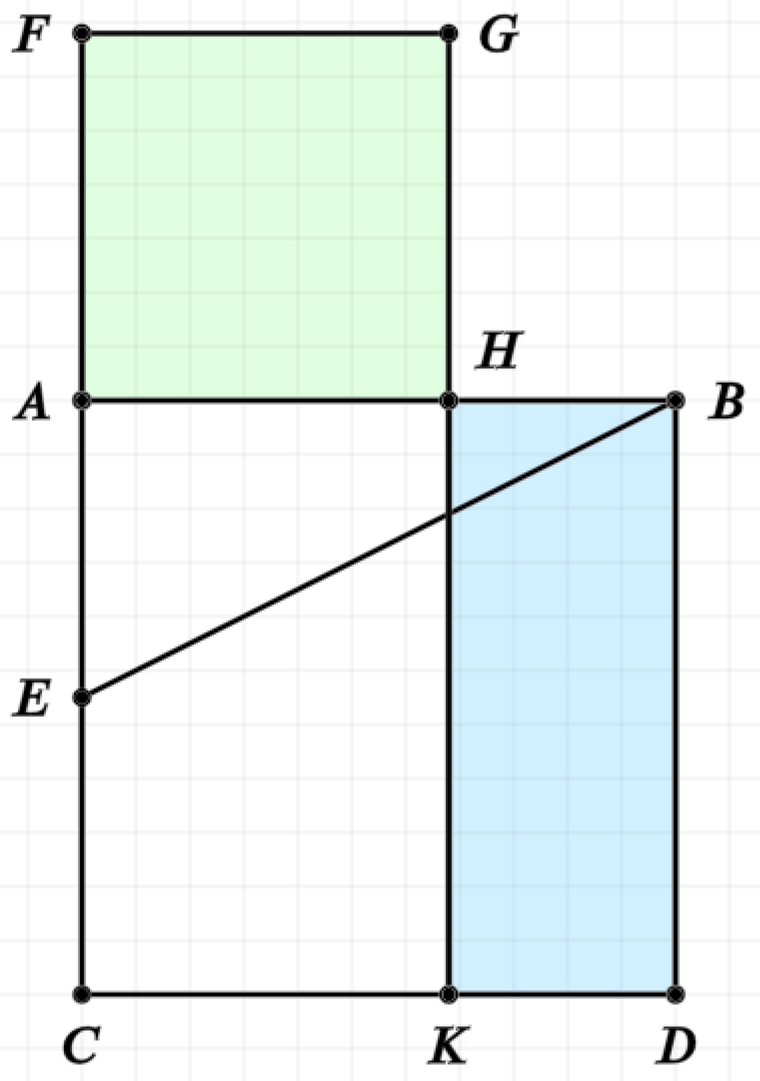
Draw a square ABCD on AB (I.46) and bisect AC (I.10) at point E

Let EB be joined, and extend CA to F such that EF equals AB

Draw a square FAGH on FA, and extend GH to line CD at point K

Proposition 11 of Book 2

To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.



$$AE = EC$$
$$EF = EB$$
$$FA = AH$$
$$FH = HD$$
$$AB \cdot BH = AH \cdot AH$$

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

Construction

Draw a square ABCD on AB (I.46) and bisect AC (I.10) at point E

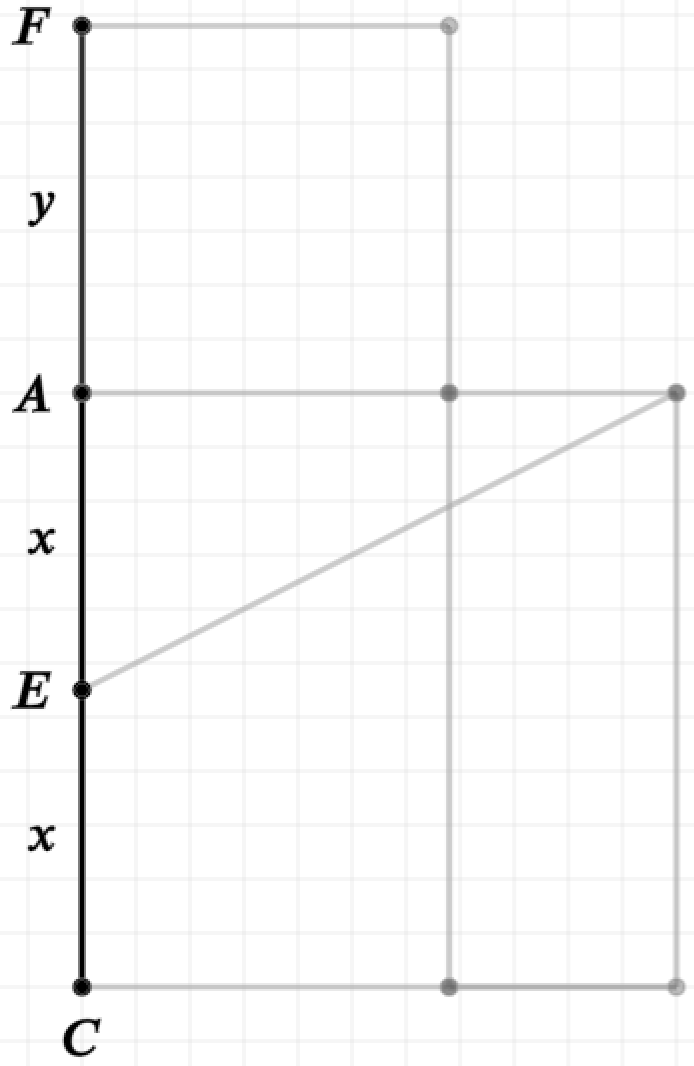
Let EB be joined, and extend CA to F such that EF equals AB

Draw a square FAGH on FA, and extend GH to line CD at point K

The point H has been defined such that FH equals HD

Proposition 11 of Book 2

To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.



$AE = EC$
$EF = EB$
$FA = AH$

$$CF \cdot AF + AE^2 = EF^2$$

$$(2x + y)\, y + x^2 = (x + y)^2$$

In other words

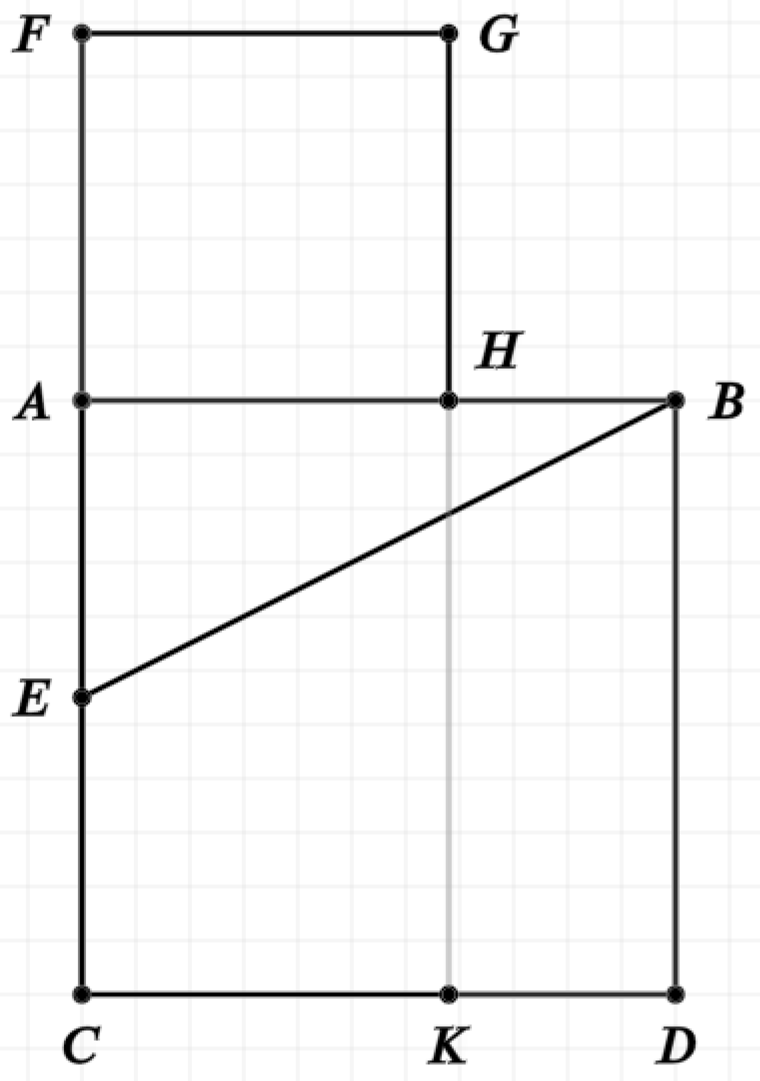
Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

Proof

From proposition 6 (II.6), if we have a bisected line, and an addition to that line, then the extended line CF times the extension AF plus the square on AE is equal to the square on EF

Proposition 11 of Book 2

To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.



$AE = EC$

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$CF \cdot AF + AE^2 = EF^2$

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In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

Proof

From proposition 6 (II.6), if we have a bisected line, and an addition to that line, then the extended line CF times the extension AF plus the square on AE is equal to the square on EF

But EB equals EF

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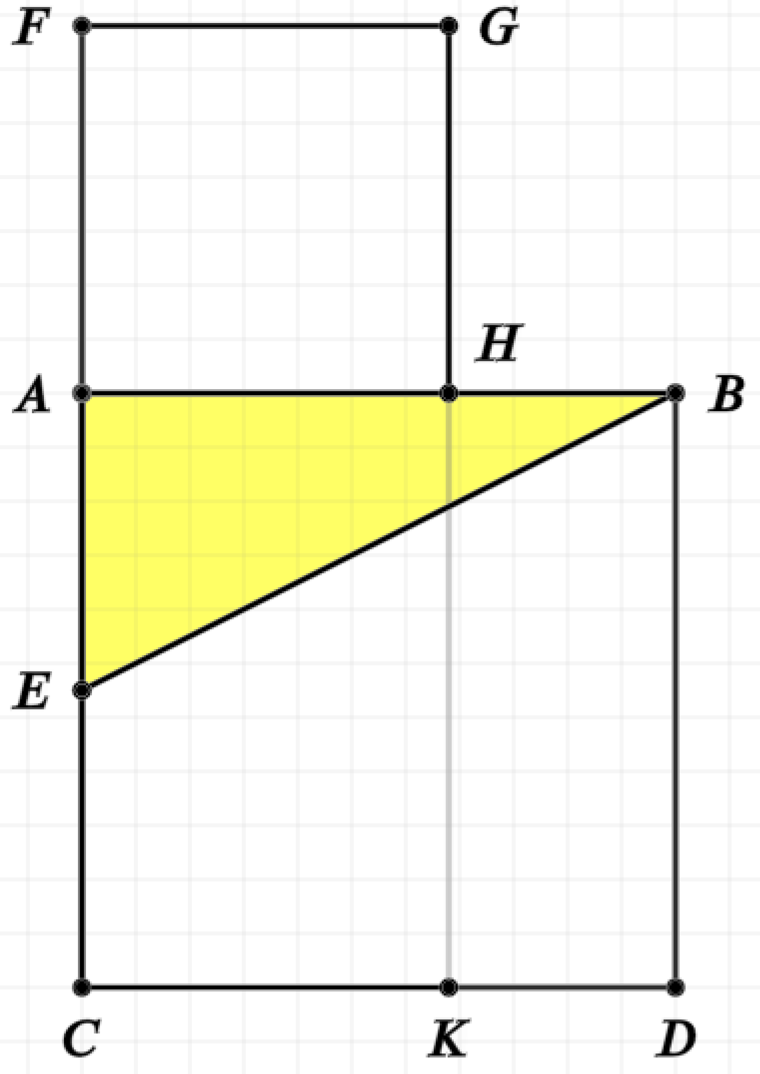
To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.



$$AE = EC$$
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$$CF \cdot AF + AE^2 = EF^2$$
$$CF \cdot AF + AE^2 = EB^2$$
$$AB^2 + AE^2 = EB^2$$

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

Proof

From proposition 6 (II.6), if we have a bisected line, and an addition to that line, then the extended line CF times the extension AF plus the square on AE is equal to the square on EF

But EB equals EF

Triangle AEB is right angle triangle, thus the square on AB plus the square on AE equals the square on EB (I.47)

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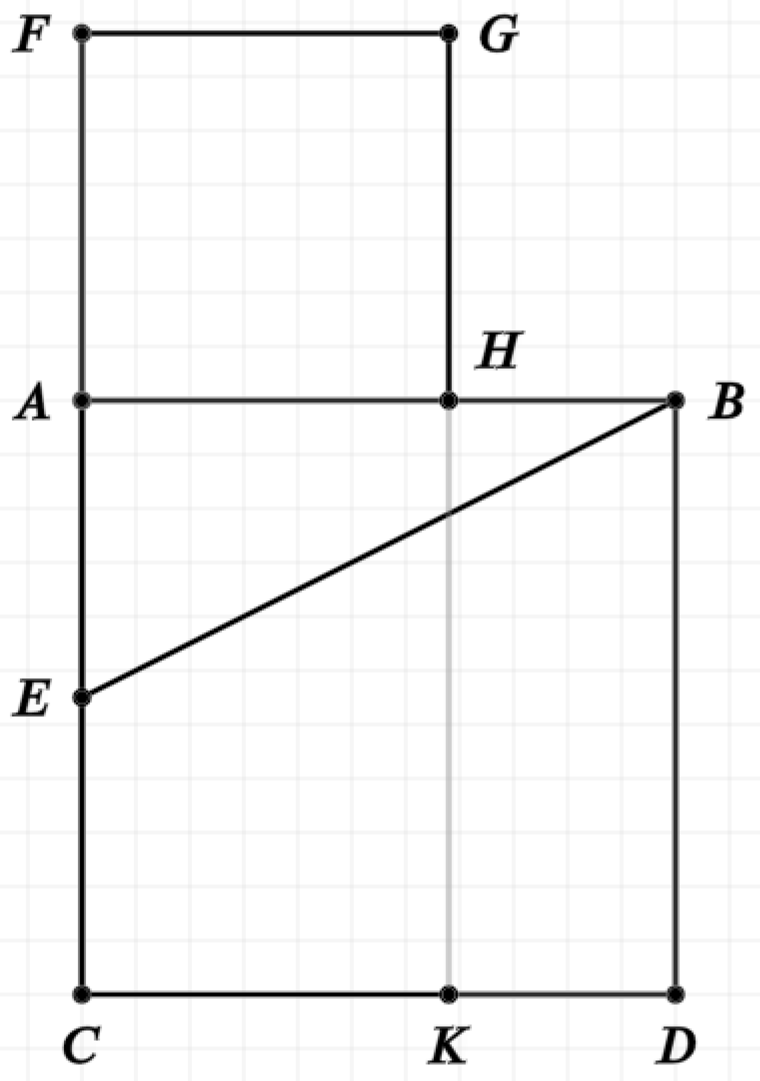

$$AE = EC$$
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$$AB^2 + AE^2 = EB^2$$
$$AB^2 = CF \cdot AF$$

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

Proof

From proposition 6 (II.6), if we have a bisected line, and an addition to that line, then the extended line CF times the extension AF plus the square on AE is equal to the square on EF

But EB equals EF

Triangle AEB is right angle triangle, thus the square on AB plus the square on AE equals the square on EB (I.47)

By comparing the equalities, we see that the square of AB is equal to the rectangle formed by CF and AF

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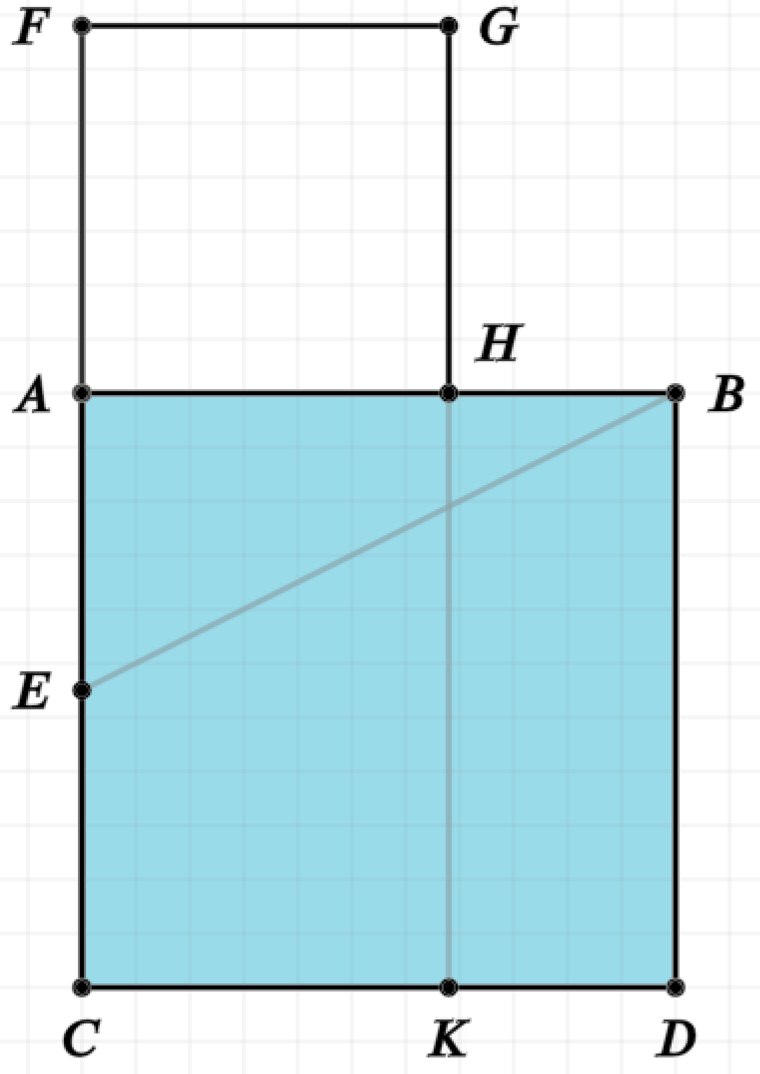
To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.



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$$CF \cdot AF + AE^2 = EF^2$$
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$$AB^2 + AE^2 = EB^2$$
$$AB^2 = CF \cdot AF$$
$$AB^2 = \square AD$$

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

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The square of AB is the rectangle AD

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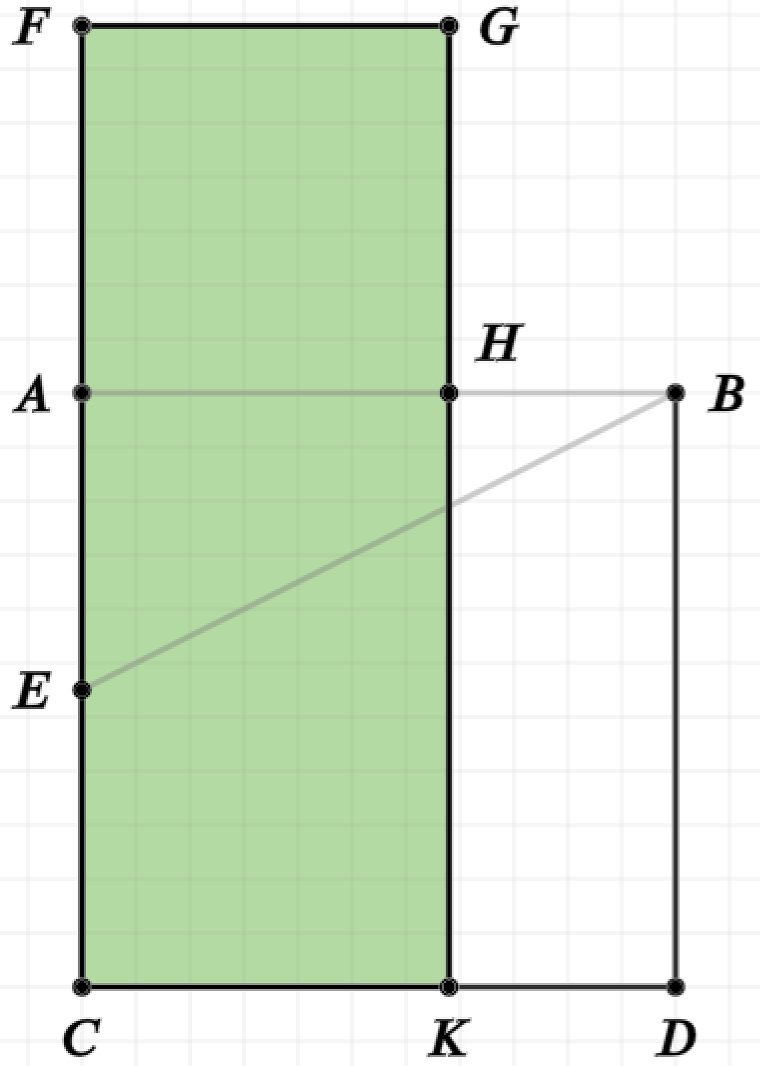

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$CF \cdot AF + AE^2 = EF^2$

$CF \cdot AF + AE^2 = EB^2$

$AB^2 + AE^2 = EB^2$

$AB^2 = CF \cdot AF$

$AB^2 = \square AD$

$CF \cdot AF = \square FK$

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The rectangle CF,AF is the rectangle FK, since AF equal AH

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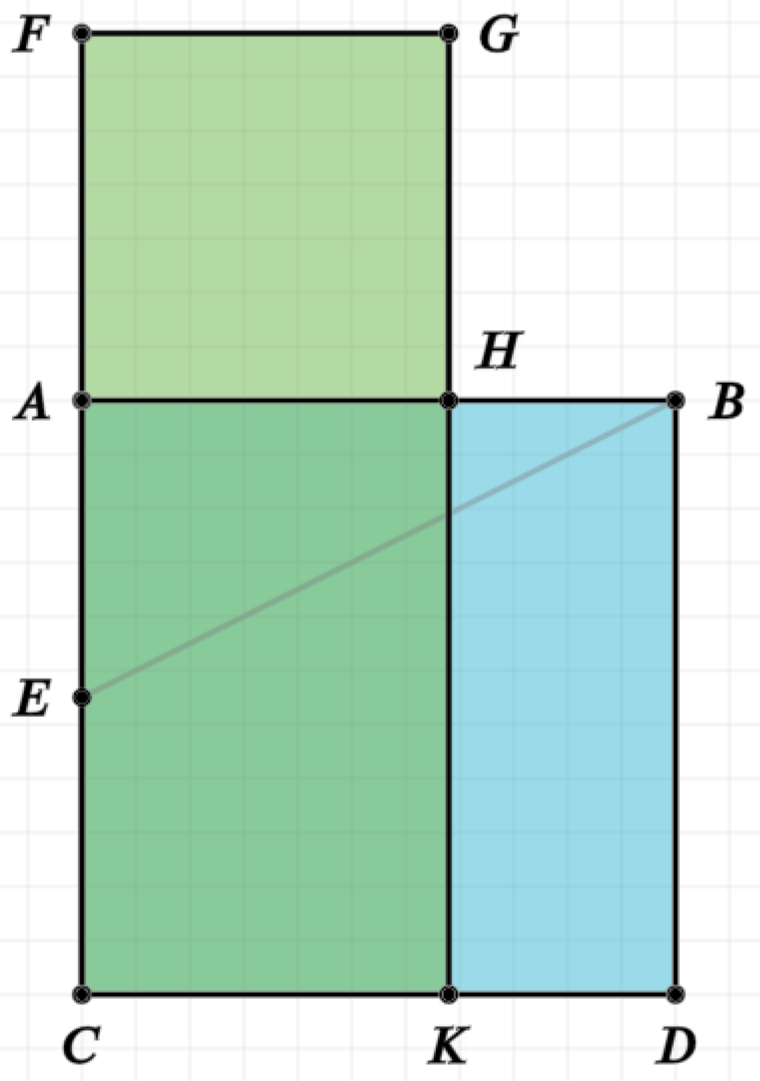

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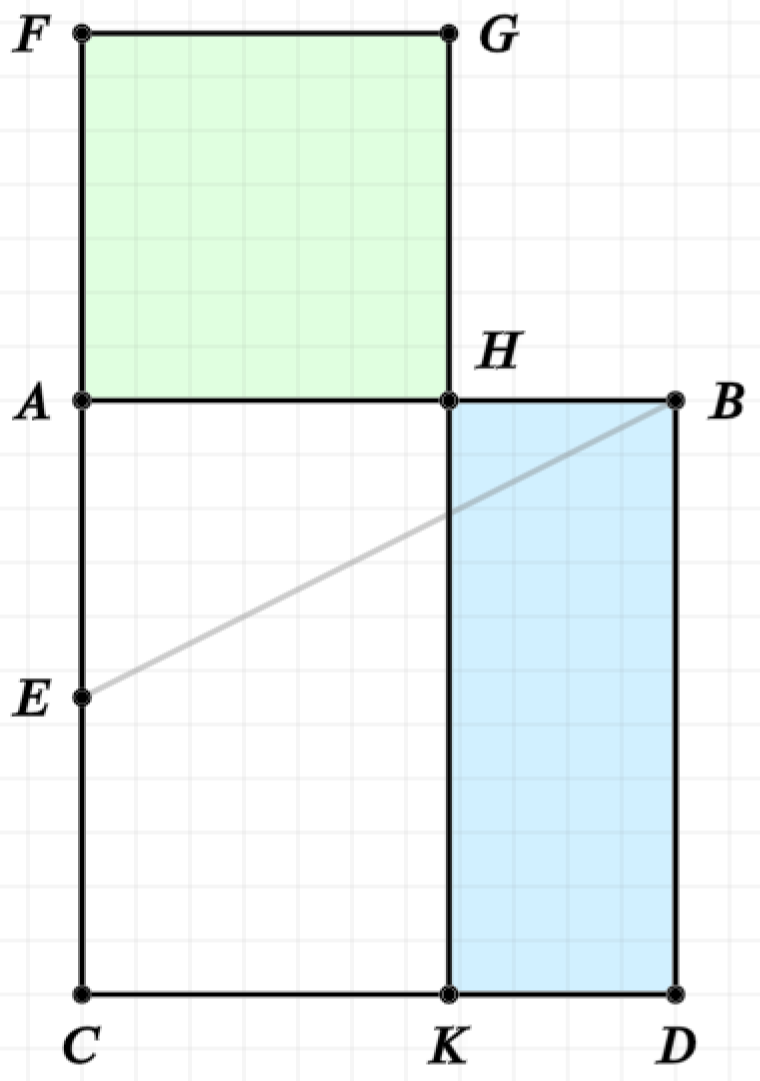

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The square of AB is the rectangle AD

The rectangle CF,AF is the rectangle FK, since AF equal AH

Subtract AK from both sides of the quality, and FH equals HD

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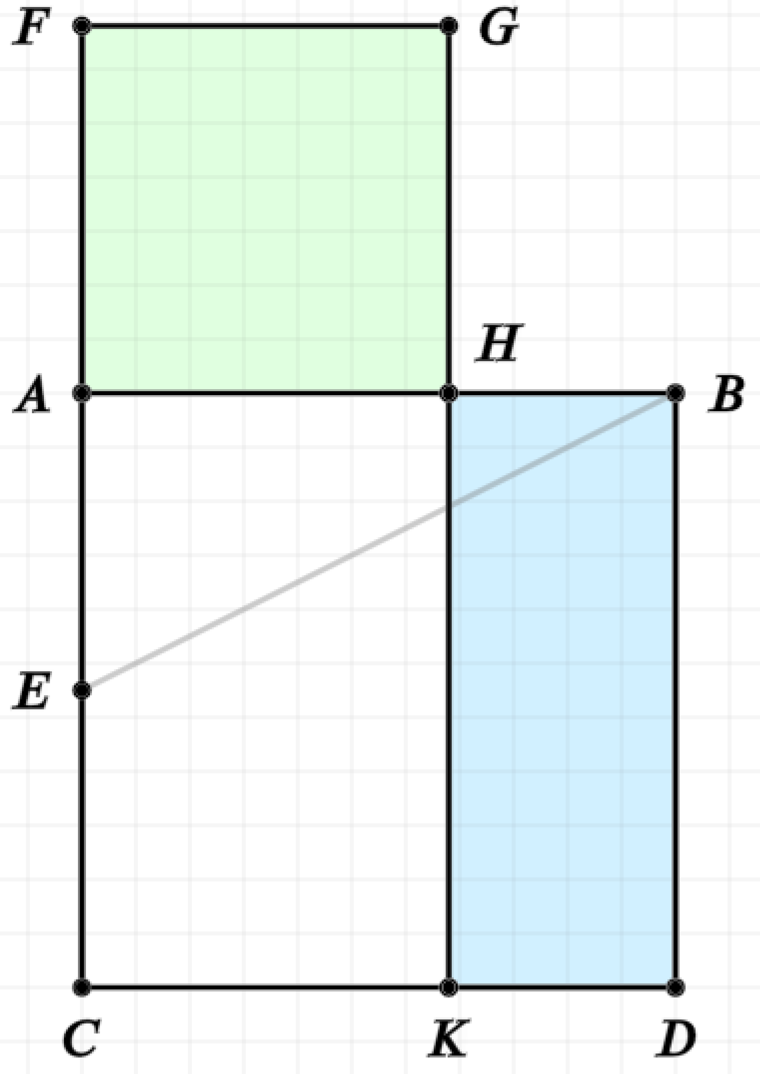

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$$AB^2 = \square AD$$
$$CF \cdot AF = \square FK$$
$$\square AD = \square FK$$
$$\square FH = \square HD$$
$$AH \cdot AH = AB \cdot BH$$

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The square of AB is the rectangle AD

The rectangle CF,AF is the rectangle FK, since AF equal AH

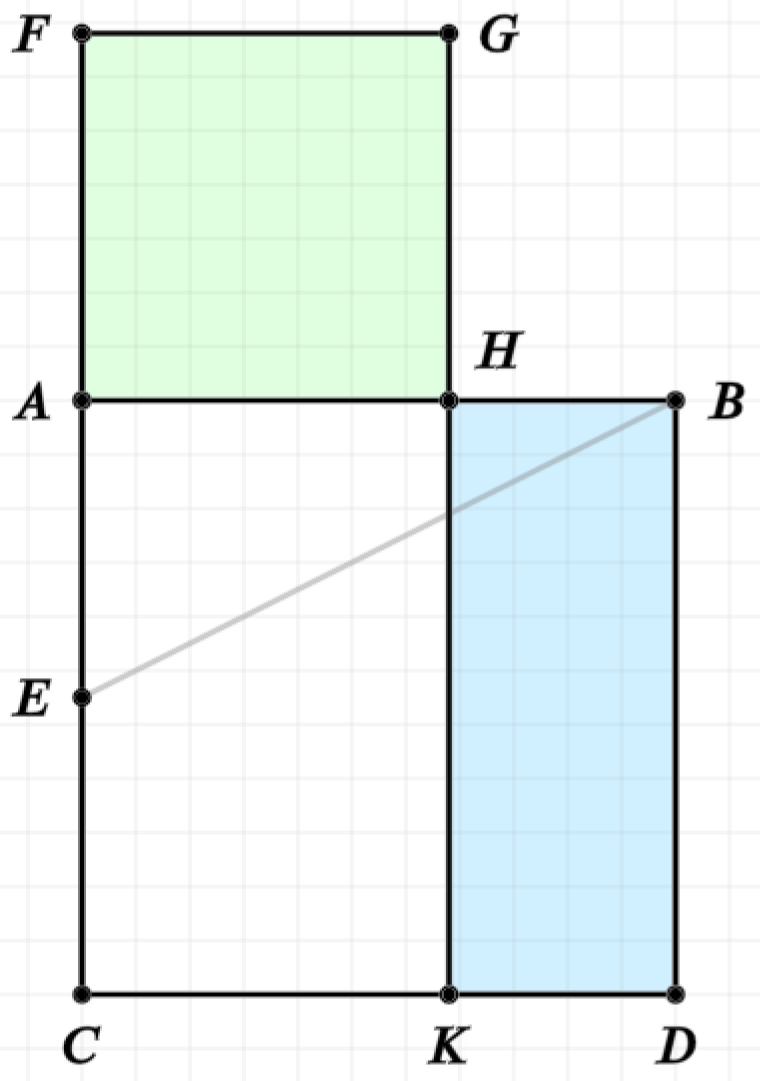
Subtract AK from both sides of the quality, and FH equals HD

But FH is formed as the square on AH, and HD is the rectangle formed by AB,BH since AB equals BD

Thus AH squared is equal to AB times BH

Proposition 11 of Book 2

To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.



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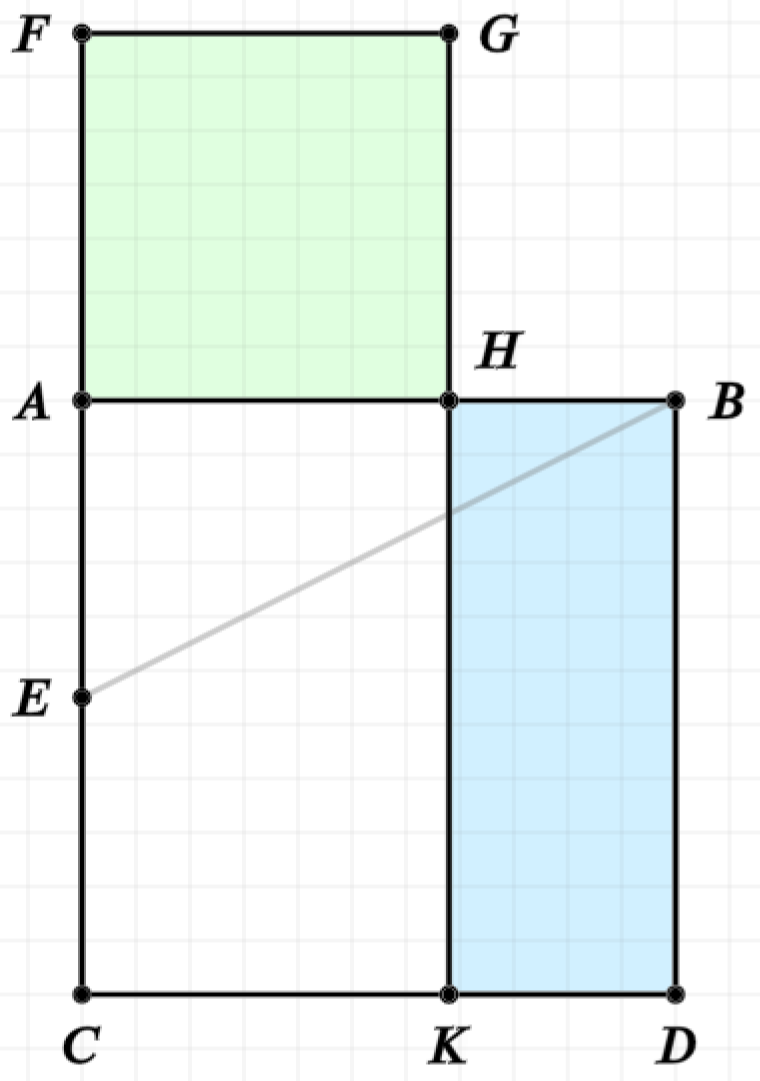

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$AB^2 = \square AD$

$CF \cdot AF = \square FK$

$\square AD = \square FK$

$\square FH = \square HD$

$AH \cdot AH = AB \cdot BH$

In other words

Find the point H on the line AB such that the rectangle formed by AB and BH is equal to the square on AH

Golden Ratio

The golden ratio is defined as:

$$\frac{a}{b} = \frac{a+b}{b}, \quad \text{where} \quad a > b$$

Since AB is equal to AH + AB, this proposition find H such that

$$AH \cdot AH = (AH + BH) \cdot BH$$

or, the golden ratio...

$$\frac{AH}{BH} = \frac{AH + BH}{AH}$$

Content:

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PDFs: https://github.com/sandy-bultena/Euclid/tree/python-manim/Propositions_PDFs

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